

Java OOP & Collections - Quick Revision Cheat Sheet

QUICK REVISION & EXAM CHEAT SHEET

Subject: Java Programming | Chapter: Chapter 2: Object-Oriented Programming & Collections | Academic Year: 2026-2027

OOP Quick Summary

- * Encapsulation: Private fields + public getters/setters.
- * Inheritance: ``class Dog extends Animal``. ``super()`` calls parent constructor.
- * Polymorphism: Parent reference pointing to child object: ``Animal a = new Dog(); a.makeSound();``.
- * Abstraction: ``abstract class`` for partial implementation; ``interface`` for pure contracts.

Collections Framework Complexity Cheat Sheet

- * ArrayList: $O(1)$ random access, $O(n)$ arbitrary insertion/deletion, dynamic growth by 50%.
- * LinkedList: $O(n)$ search, $O(1)$ insertion at ends, higher memory overhead per node.
- * HashSet: $O(1)$ lookup, unordered, relies on `equals()` and `hashCode()` contract.
- * TreeSet / TreeMap: $O(\log n)$ lookup, sorted by `Comparable` or `Comparator`.
- * HashMap: Key-value pairs, $O(1)$ average lookup, converts buckets to red-black trees at threshold 8.

Equals and hashCode Contract

- * If two objects are equal according to `equals()`, their `hashCode()` MUST be identical.
- * If two objects have the same `hashCode()`, they are not necessarily equal (hash collision).
- * Always override both `equals()` and `hashCode()` together when using custom keys in HashMaps or HashSets.